QC STORY PROJECT

PROJECT TITLE :- Reduction in back door D pillar bolt slippage by 80% at FRC For all Scorpio Models

Mahindra and Mahindra Ltd, Plant I, Scorpio Body shop.

	Plan	Actual
Start Date	**29.06.12**	**25.7.12**
Completion Date	**22.08.12**	**06.10.12**

Current Status:

Completed

(Reduction in bolt slippage by 87%)

Cross Functional Team

No	Description	Name	Designation	Dept	Phone	Email
1	Team Leader	Dnyaneshwar Vispute	Astt Mgr	Prodn	8888807375	DNAYNESHWAR.VISPUTE@mahindra.com
2	Mentor	R.G.Suryavanshi	DGM	Head	9881378988	SURYAVANSHI.RAVINDRA@mahindra.com
3	Member 1	Vivek Bhat	Dy. Mgr	Tooling		BHAT.VIVEK@mahindra.com
4	Member 2	Manoj More	Dy. Mgr	Prodn		MORE.MANOJ@mahindra.com
5	Member 3	S.Y.Deshpande	Manager	QA		DESHPANDE.SANDEEP2@mahindra.com
6	Member 4	R.A.Sawant	Manager	Prodn		SAWANT.RAJENDRA2@mahindra.com
7	Member 5	A.S Hasme	Manager	Tooling		HASME.ASHUTOSH@mahindra.com

1. Select the Theme

Brief description of Problem

What: Bolt Slippage on RH D pillar back door mounting

Where: Back Door fitment stage

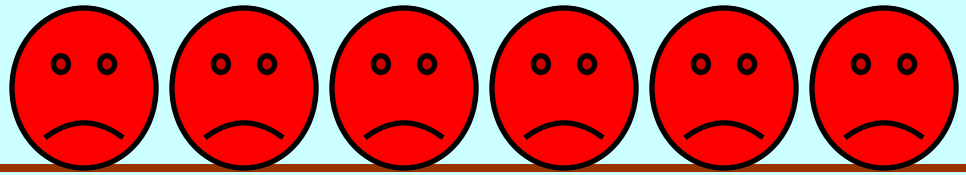
Who: Operator ,check man ,finisher and buy off operator.

When: Daily (All shifts)

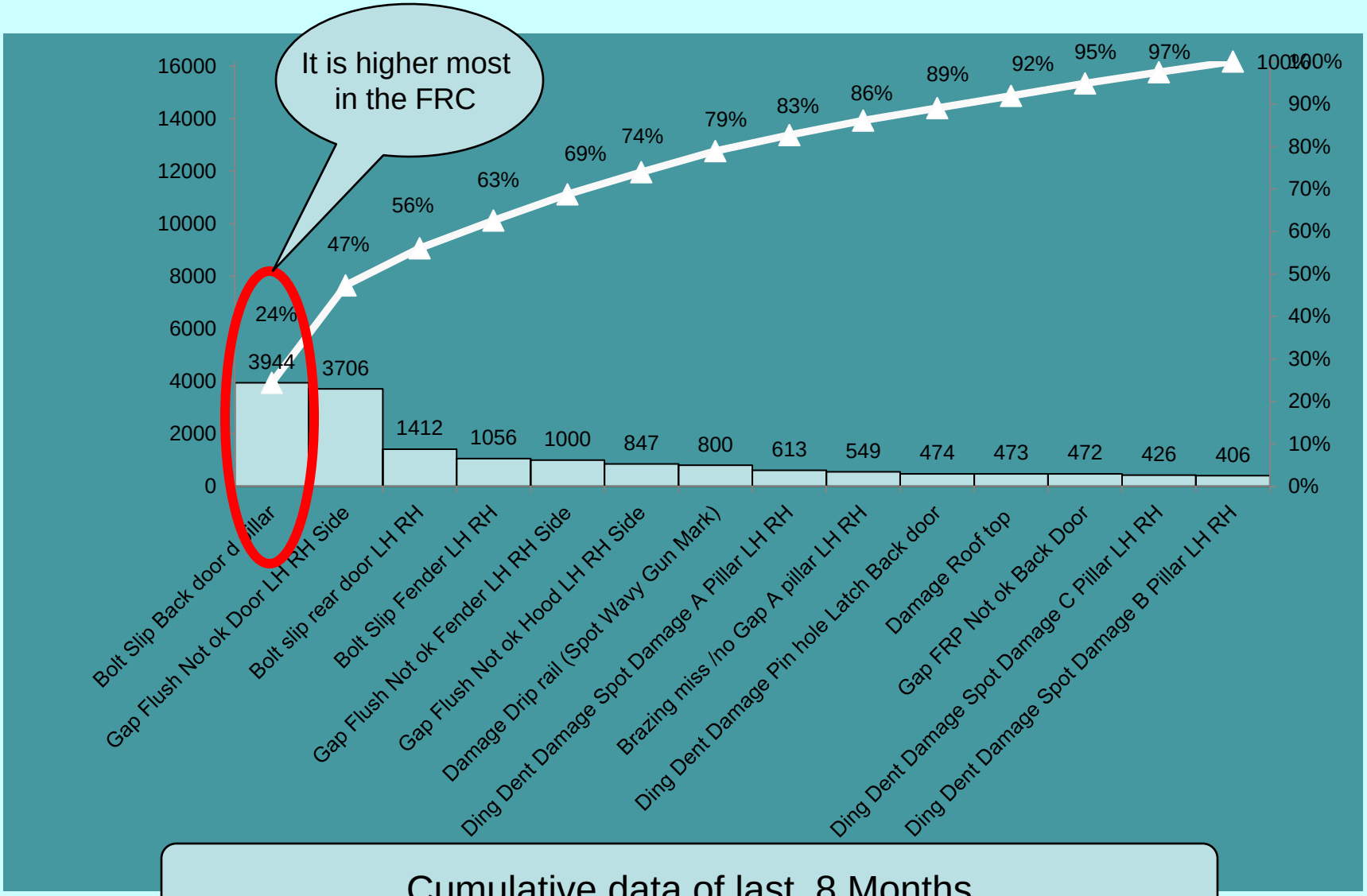
Why: COPQ of the defect is Rs 3.13 Lac per Year

How: Visual appearance at buy off

Since more than 1 years we are struggling with this chronic defect

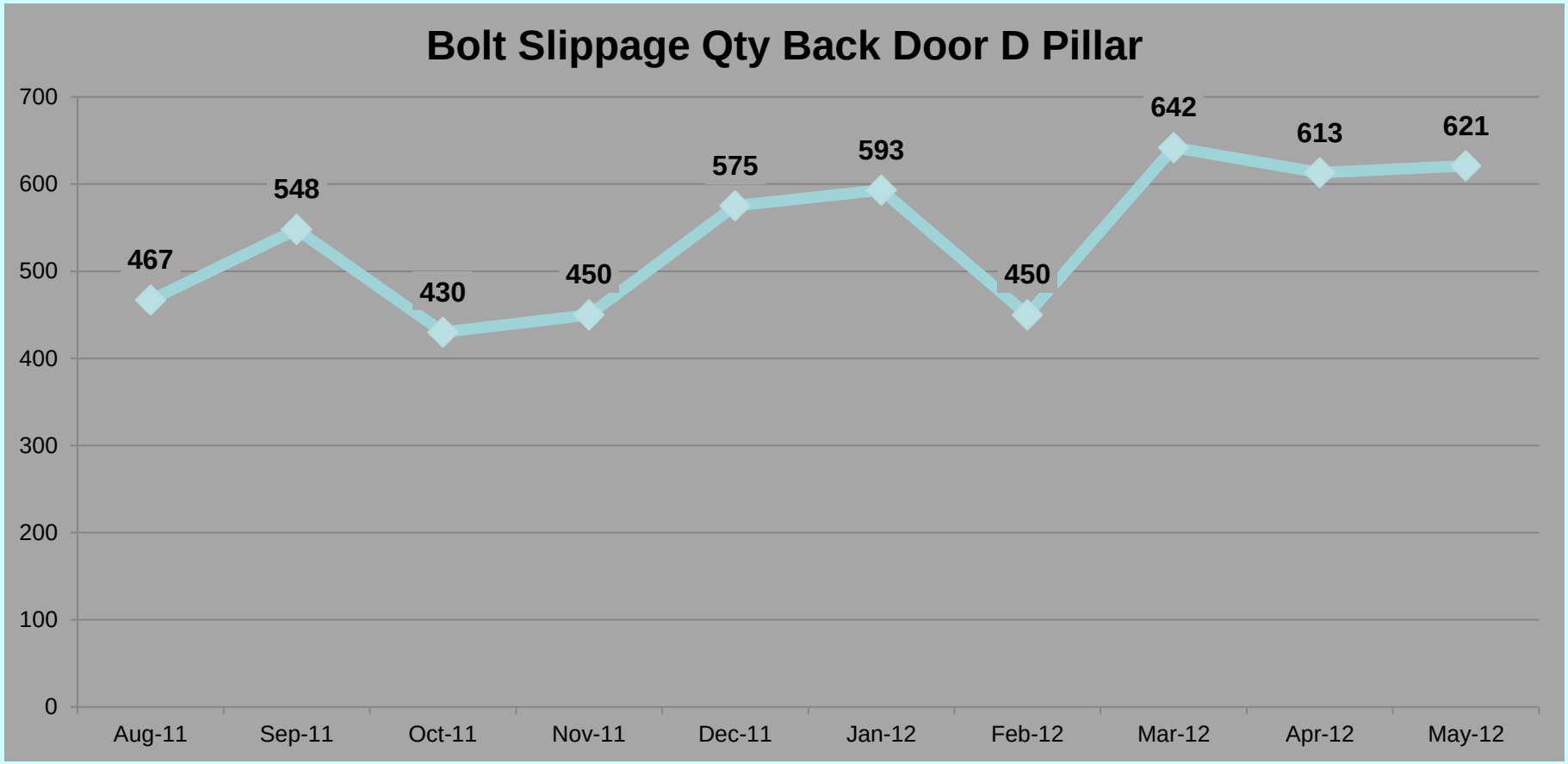


2. Justify the choice (reasons)



Cumulative data of last 8 Months

2. Justify the choice (reasons)



Month wise defect trend F-12

2. Justify the choice (reasons)

D. COPQ (Cost of Poor Quality) Calculation

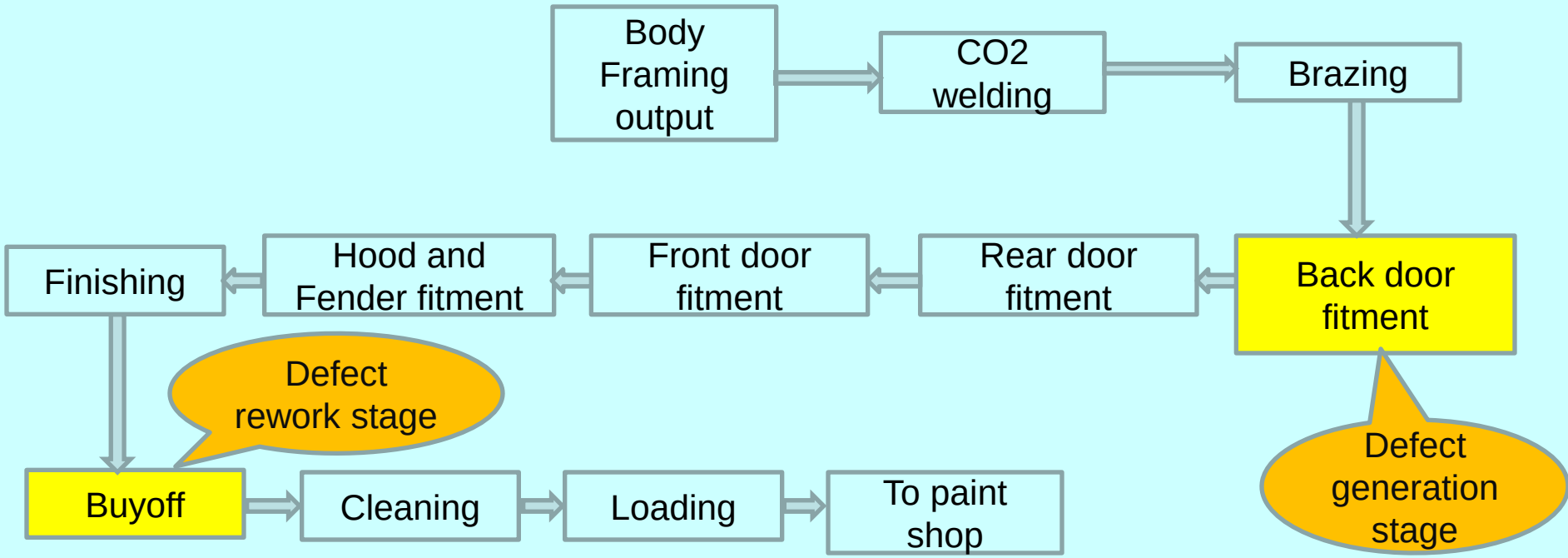
Description	Avg Per Month Bolt slippage
Number of pieces scrapped avg per month (Allen Head M10)	493
Number of pieces(Vehicle) reworked last month	493
Part cost/piece	9
Rework cost/piece(Vehicle) Including (Labor Cost)	44
Total scrap cost (Rs.) for last month	4437
Total rework cost(Rs.) for last month	21659
Total Rejection cost (Rs.) for last month (Part Cost + Total rework cost)	26096
Annual rejection and scrap cost of the defect in Rs	313153

2.a Immediate Containment Action

Initiated 100 % checking and rework of the defective BIW at buy off

3. Understand the current situation

Observation of process

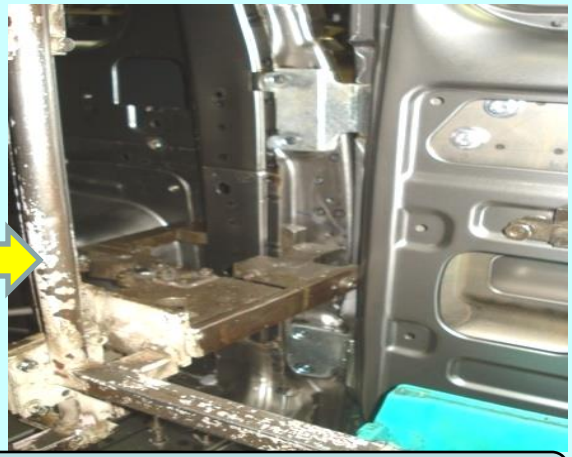


3. Understand the current situation

SOP – BACK DOOR FITMENT

Back door locating in BIW

Locking of the back door fixture in BIW With manual clamps



Fixture BIW
locating lower clamp

Tightening of the bolts on BIW d pillar

Release and removal of fixture from BIW

4. Select Target

Present Level:

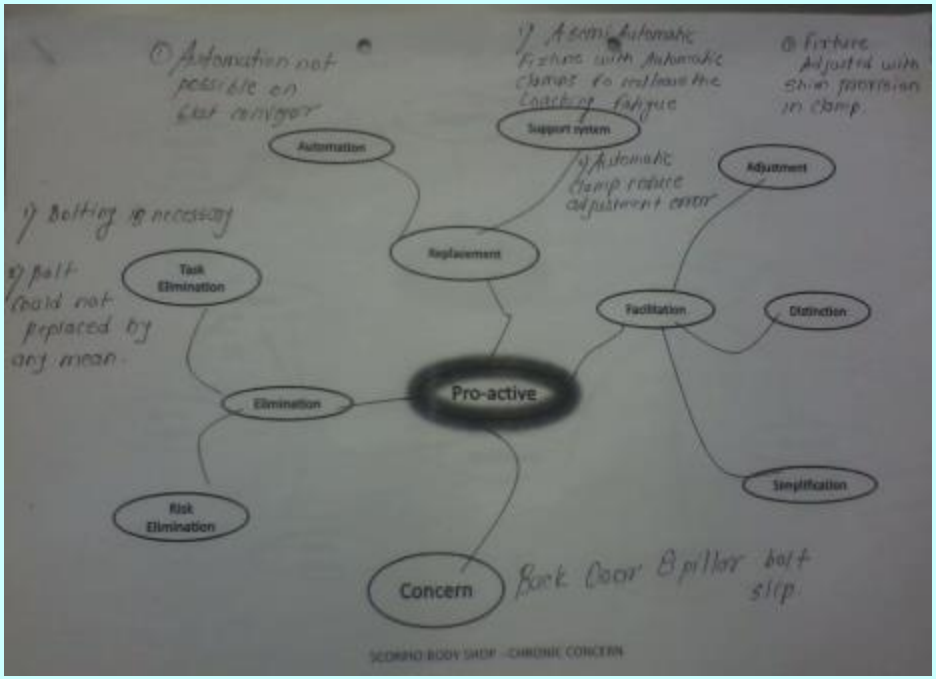
Avg. 493 BIW per month at BUY OFF

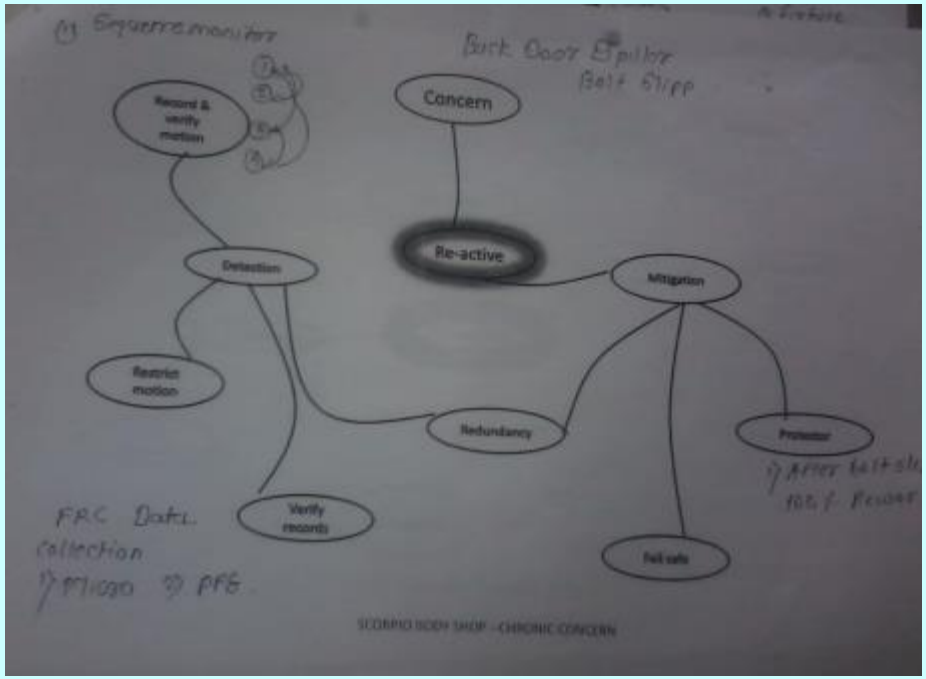
Target:

FRC Level :80% REDUCTION of defect of Back door bolt slippage by Oct 12 (Avg . 66 BIW per month)

5. Analysis

5.1 Brain Mapping:





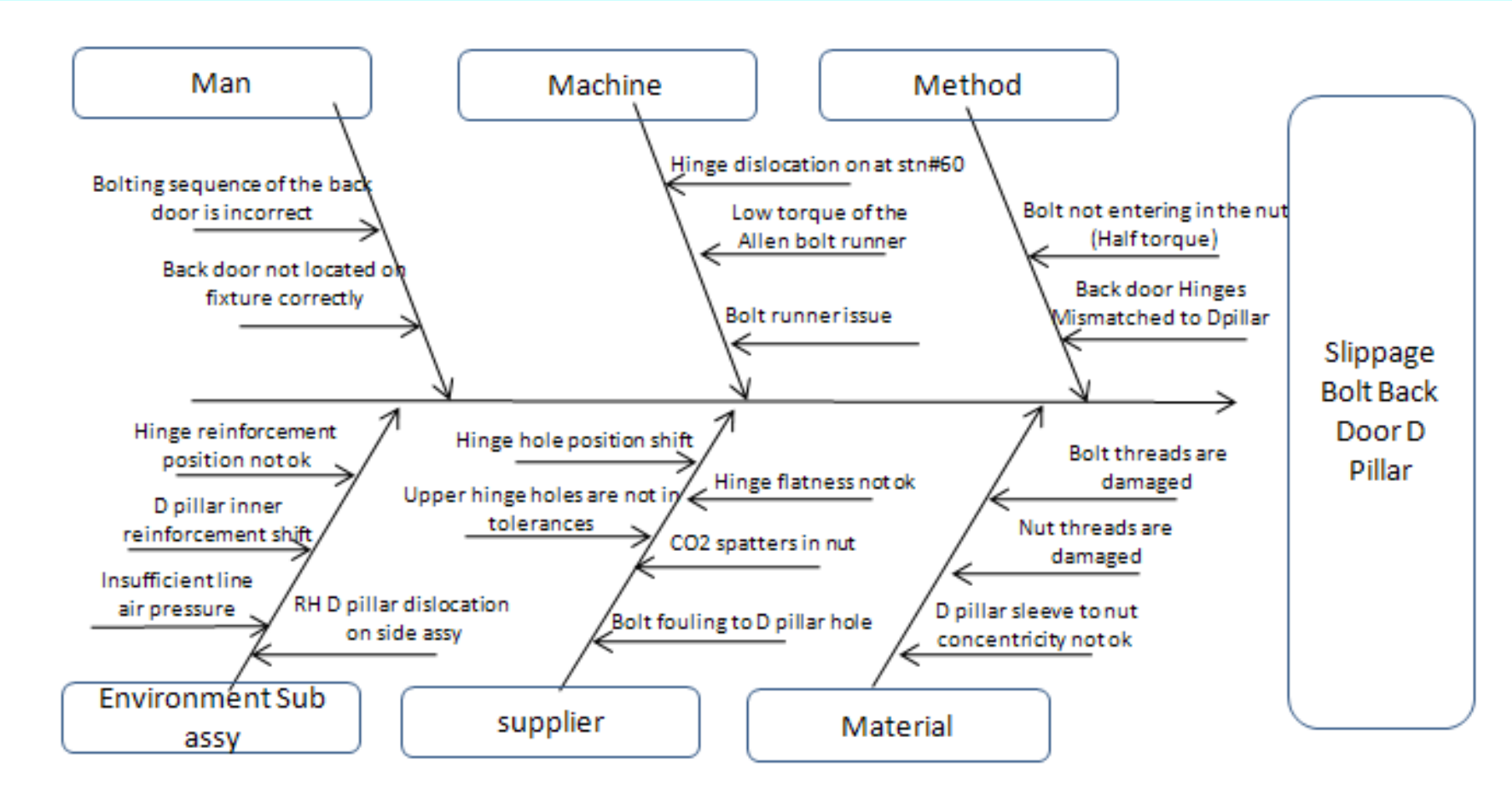
5. Analysis

5.2 Identify possible causes:

Sr No	All Possible Cause (CFT- BRAINSTORMING)
1	Back door hinges mismatched to the D pillar holes
2	Bolt not entering in the nut (Half torque)
3	Bolt fouling to D pillar hole
4	Low torque of the Allen bolt runner
5	Bolting sequence of the back door is incorrect
6	Bolt threads are damaged
7	Upper hinge holes are not in tolerances
8	Back door not located on fixture correctly
9	Hinge dislocation on at stn#60
10	Hinge hole position shift
11	Hinge flatness not ok
12	Hinge reinforcement position not ok
13	Bolt runner issue
14	Insufficient line air pressure
15	D pillar inner reinforcement shift
16	D pillar sleeve to nut concentricity not ok
17	RH D pillar dislocation on side assy

Cause Verification

5.2 CAUSE AND EFFECT DIAGRAME



5. Analysis

5.2 Confirm the probable cause:

Sr No	Probable Causes
1	Back door hinges mismatched to the D pillar holes
2	Low torque of the Allen bolt runner
3	Bolting sequence of the back door is incorrect
4	Bolt threads are damaged
5	Upper hinge holes are not in tolerances
6	Back door not located on fixture correctly
7	Hinge dislocation on at stn#60
8	Hinge reinforcement position not ok
9	D pillar inner reinforcement shift
10	D pillar sleeve to nut concentricity not ok
11	RH D pillar dislocation on side assy
12	CO2 Spatters in nut

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
1.	**Back door hinges mismatched to D pillar holes**	Trail taken for the 50 BIW found hinges getting shift and misalign during setting fixture locking to BIW	**Valid**

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
2.	**Low torque of the Allen bolt runner**	Daily checking of the runner in ACTA Audit	**In valid**

Date	1-Jun	2-Jun	3-Jun	4-Jun	5-Jun	6-Jun	7-Jun	8-Jun	9-Jun	10-Jun	11-Jun	12-Jun	13-Jun	14-Jun	15-Jun
1	54								51	54	57	54	53.62	57	56
2	56								56	55	53	56	55.01	56	58
3	57								53	54	55	57	54.23	54	54
4	55								53	56	58	56	53.68	55	56
X bar	55.50								53.25	54.75	55.75	55.75	54.14	55.50	56.00
R bar	3								5	2	5	3	1.39	3	4
UCL x bar	56.15														
LCL x bar	51.48														
UCL r bar	7.29														
LCL r bar	0														

16-Jun	17-Jun	18-Jun	19-Jun	20-Jun	21-Jun	22-Jun	23-Jun	24-Jun	25-Jun	26-Jun	27-Jun	28-Jun	29-Jun	30-Jun
		50	52	53	54	50			53	52	56			52
		52	53	53	54	52			52	55	56			54
		48	50	51	52	52			51	54	55			60
		51	52	52	53	52			52	53	54			52
		50.25	51.75	52.25	53.25	51.50			52.00	53.50	55.25			54.50
		4	3	2	2	2			2	3	2			8

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
3	**Bolting sequence of the back door is incorrect**	Change in the bolting sequence cause the bolt slippage	**Valid**

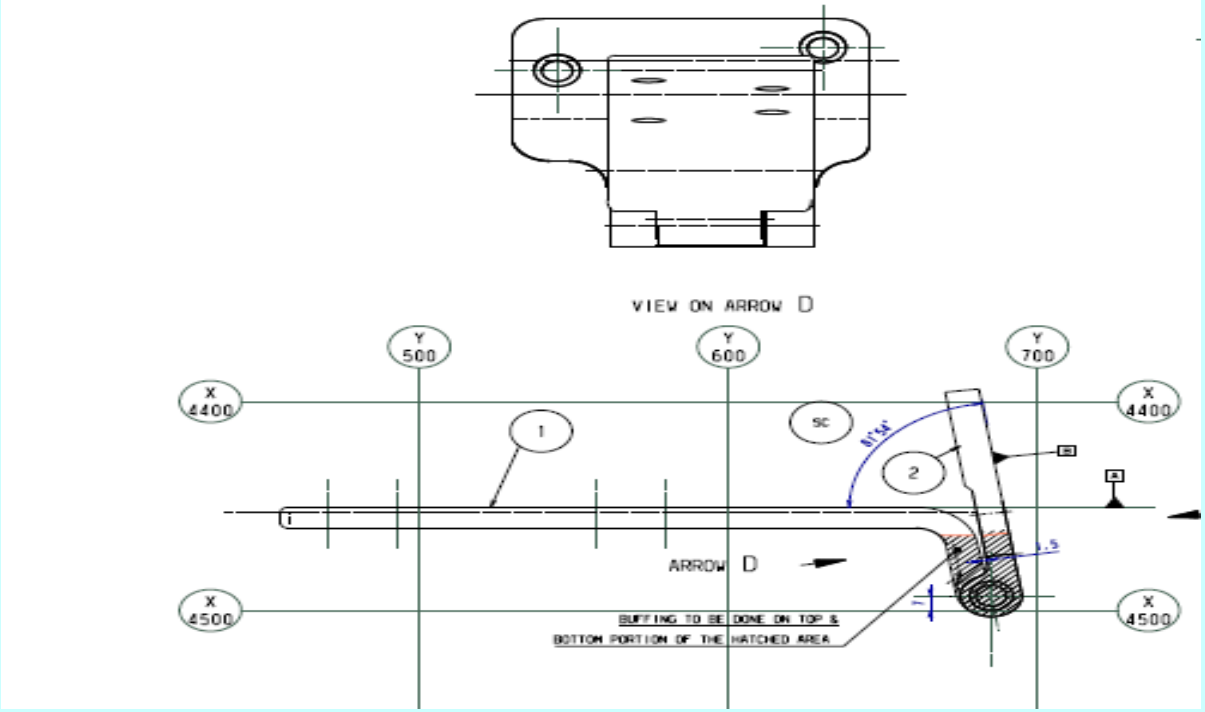
External Audit done with the third party to observe the bolting Sequence and slippage concern

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
4	**Bolt threads are damaged**	100 Bolts checked and observed the thread damaged bolts 59 having bolt slippage issue	**Valid**

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
5	**Upper hinge holes are not in tolerances**	Upper hinge not as per dimension 50Hinges checked of which 35 found NOT OK	**Valid**



5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
6	**Back door not located on fixture correctly**	35 Door fitment checked, found mismatch in 11 cases.	**Valid**
7	**Hinge reinforcement position not ok**	35 Door assy checked found no deformation and defect	**Invalid**

Hinge Assy at #50

Hinge Reinforcement

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
8	**D pillar sleeve to nut concentricity not ok**	Bolt fouling of the d pillar after Checking 50BIW 36 found defect of fouling due to no concentricity	**Valid**

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
9	**RH D pillar dislocation on side assy**	Automation si there in side assy no chance to shift D pillar	**Invalid**

5.2 Confirm the probable cause:

No	Probable Cause	Testing and Observations	Conclusion
10	CO2 Spatters in nut	Bolt fouling of the d pillar after Checking 50BIW 26 found defect of fouling due to bolt not entering in it	**Valid**

WHY - WHY Analysis

5.3 Identify the root causes:

Sr No	Probable cause	why	why	why	why
1	**Back door hinges mismatched to the D pillar holes**	Holes are of hinges are not matching with the D pillar holes	Hinges holes are shifting from the back door base	Fixture movement and clamping variation	
3	**Bolting sequence of the back door is incorrect**	Wrong bolt sequence	Operator forgot to follow the sequence	No visual assist is thee to remember the sequence	
4	**Bolt threads are damaged**	Bolt get locked in mid of the nut	Bolt thread material is get removed out and restrict the entering of it in nut	Bolt threads are damaged	
5	**Upper hinge holes are not in tolerances**	Variation in machining	Reference point not provided	No provision on drilling fixture at supplier	
6	**Back door not located in fixture correctly**	Back door not get clamped correctly during manual clamping	Reference point and location have to adjusted during manual clamping	Manual clamp reference is there	

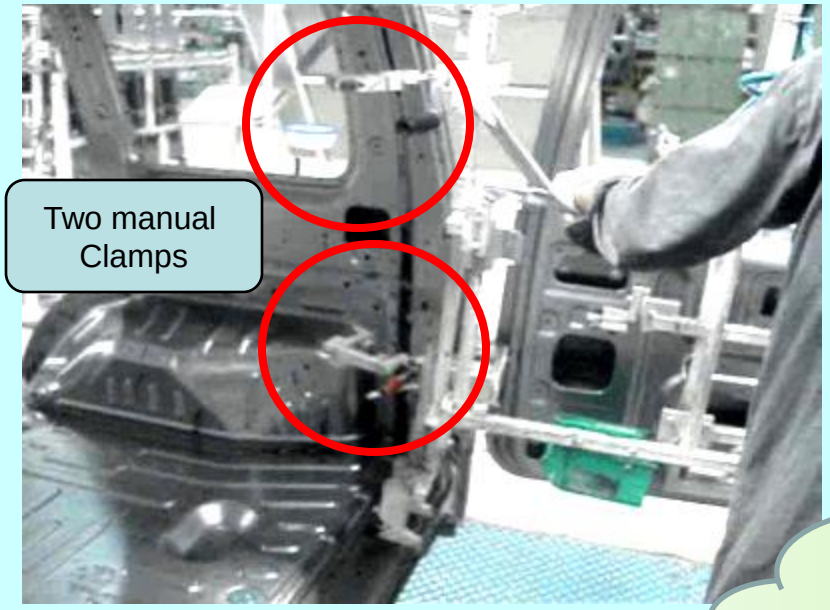
Cause Verification

5.3 Identify the root causes:

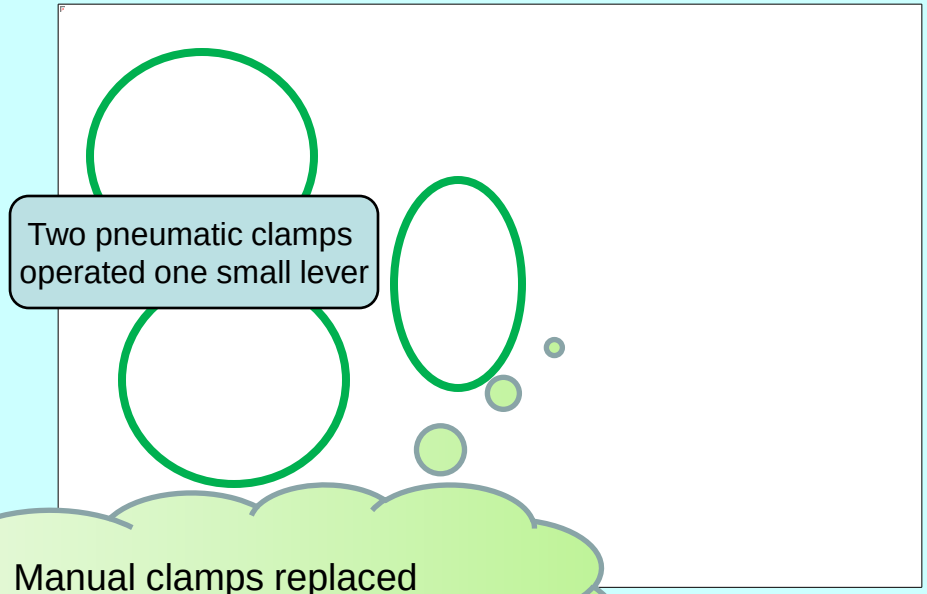
Sr No	Probable cause	why	why	why	why
8	**D pillar nut to sleeve concentricity not there**	During assy of D at supplier internal nut plate supplier is shifted	Plea in the fixture to move and shifting of nut plate	No provision to restrict shifting	
10	**Co2 spatters in the D pillar NUTs**	Co2 spatters get stick in the nut threads	During co2 welding no provision to restrict the spatters entering into the nuts	No provision on fixture	

6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
1	Back door hinges mismatched to the D pillar holes	Fixture to be modified with pneumatic clamps for the BIW CLAMPING in house to restrict the movements of hinge shifting and manual clamping	12/07/2012	Completed



Before



Manual clamps replaced with the pneumatic operated clamps

After

6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
3	Bolting sequence of the back door is incorrect	VISUAL bolt sequencing template displayed on shop floor Back Door closure station	14/08/2012	Completed

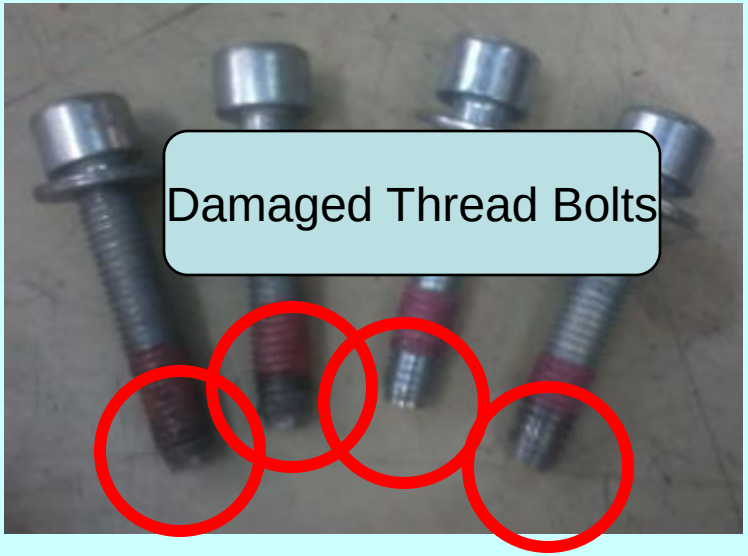
No any visuals to correct bolting sequence

Before

After

6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
4	Bolt thread are damaged	100% inspection at supplier and checked bolts to be deputed on Line	21/09/2012	Completed

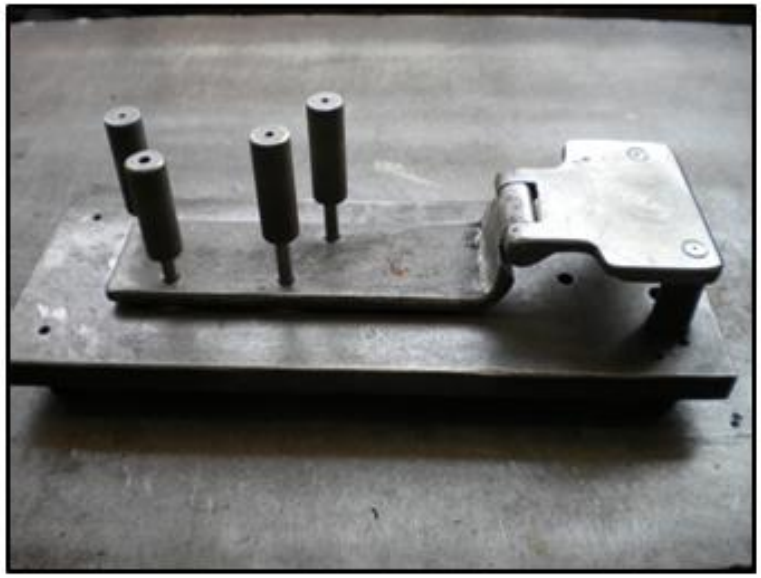


Before

After

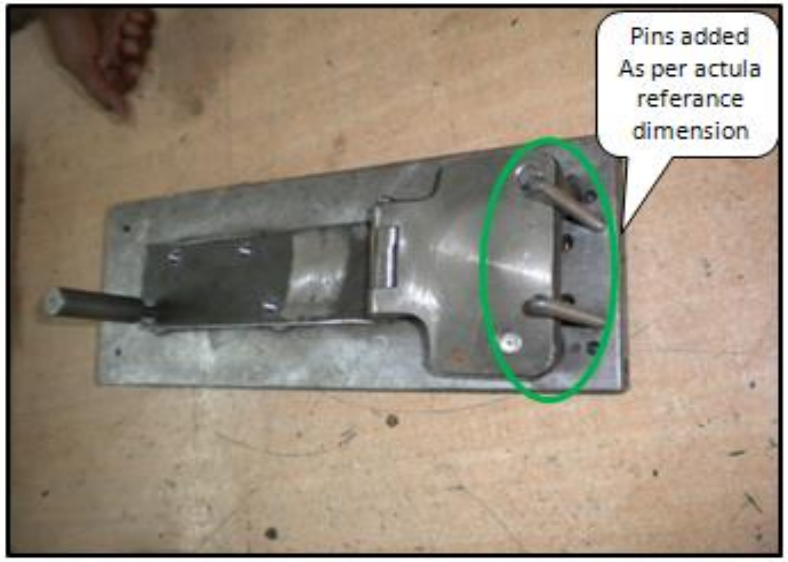
6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
5	Upper Hinge holes are not in tolerance	New drilling fixture with correct location reference pin added at supplier (Siddheshwar Forging)	21/09/2012	Completed



Initially there was no pins to lock Profile dimension of Hinge post

Before



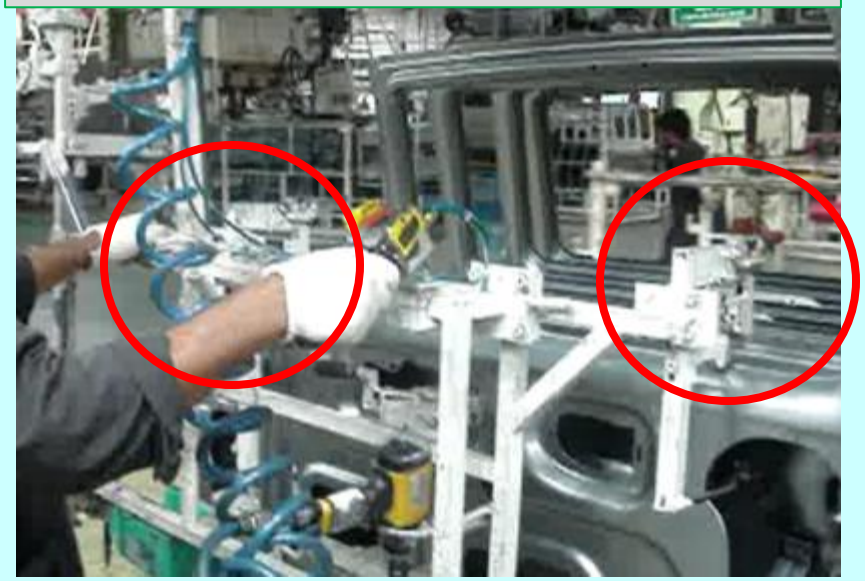
Pins are added in CD gauge to lock Profile dimension of Hinge post

After

6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
6	Back Door fixture not located in fixture correctly	Fixture to be modified with pneumatic clamps for the BACK DOOR CLAMPING in house to restrict the movements of BACK DOOR and manual clamping	12/07/2012	Completed

Two manul Back door locking clamps



Before

Single pneumatic operator Back door locking clamps

After

6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
8	D pillar nut to sleeve concentricity is not there	Stopper to be provided at D pillar sub assy stage at supplier	21/09/2012	Completed

After

6. Implement the corrective measures

Sr No	Root Causes	Action plan	Target Date	Status
10	Bolt not entering in the nut (Half torque)	To restrict the weld spatter in nut fixture to be introduced at supplier end with the nut locating CU pins 100% Retapping of nut at supplier end	18/7/2012	Completed

Co2 nut welding done directly on part

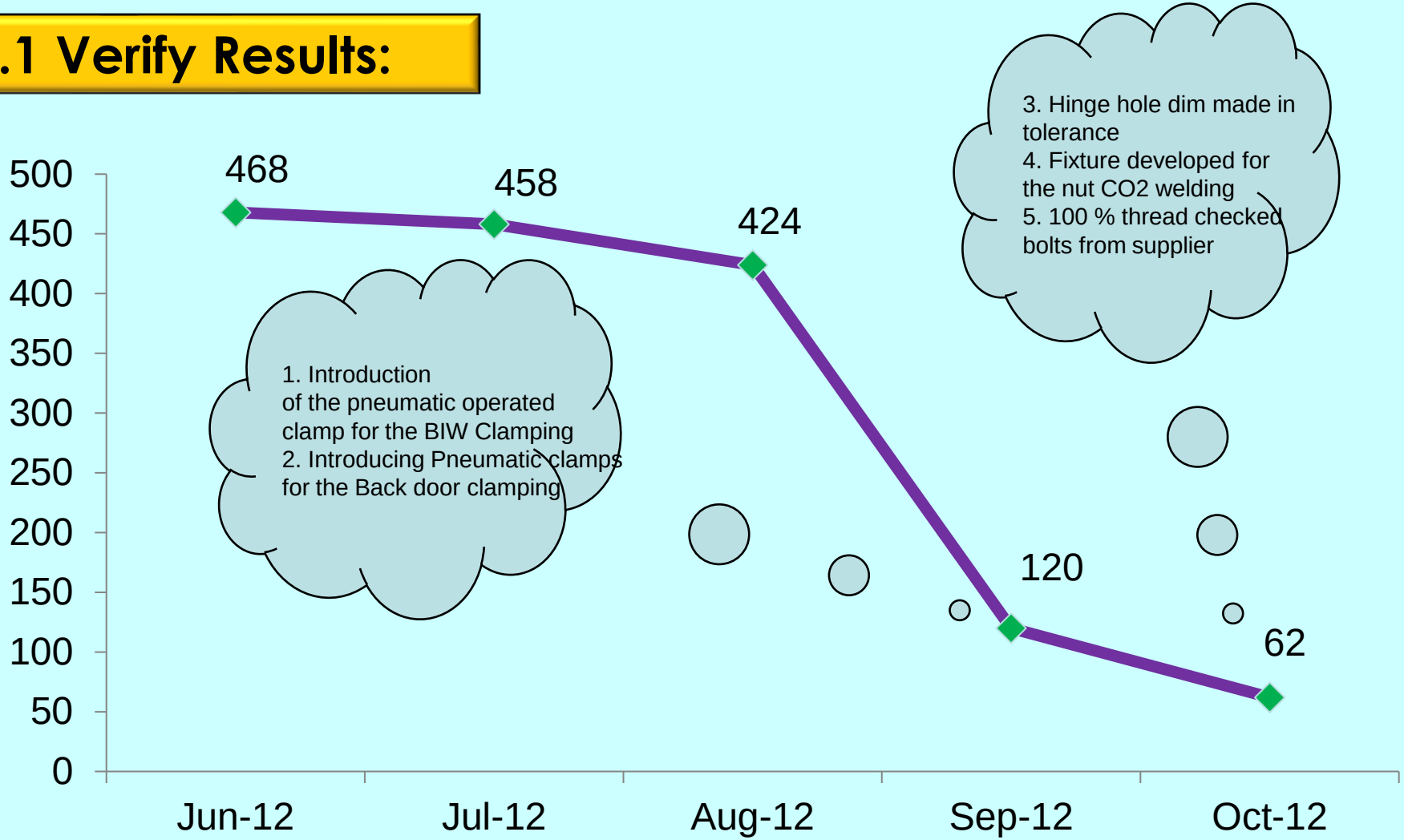
Before

Fixture is developed to avoid entry of spatters into nuts

After

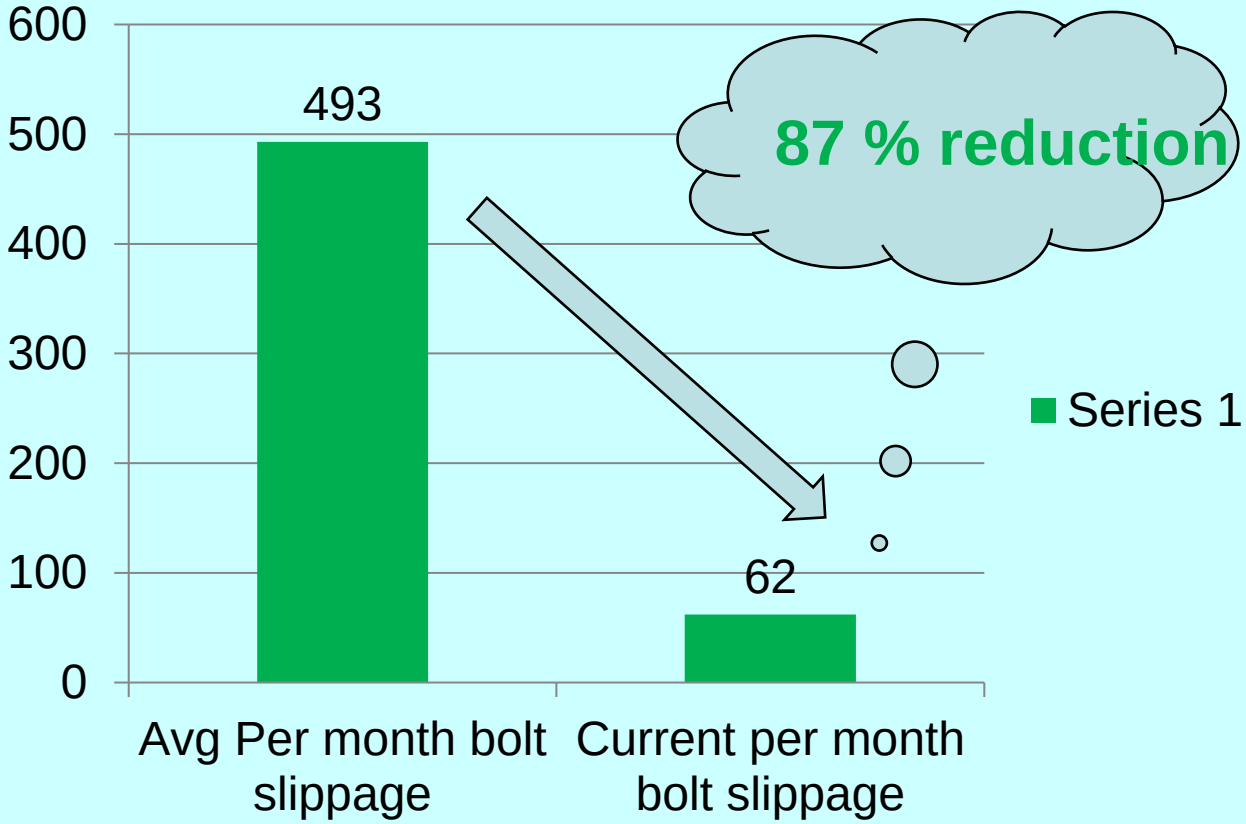
7. Check & confirm the results & effects

7.1 Verify Results:



7. Check & confirm the results & effects

7.1 Verify Results:



7.2 Full Scale Implementation plan:

We have implemented all the actions

7.Check & confirm the results & effects (Cont……)

7.3 Tangible benefits:

1.Heavy rework at in house is avoided

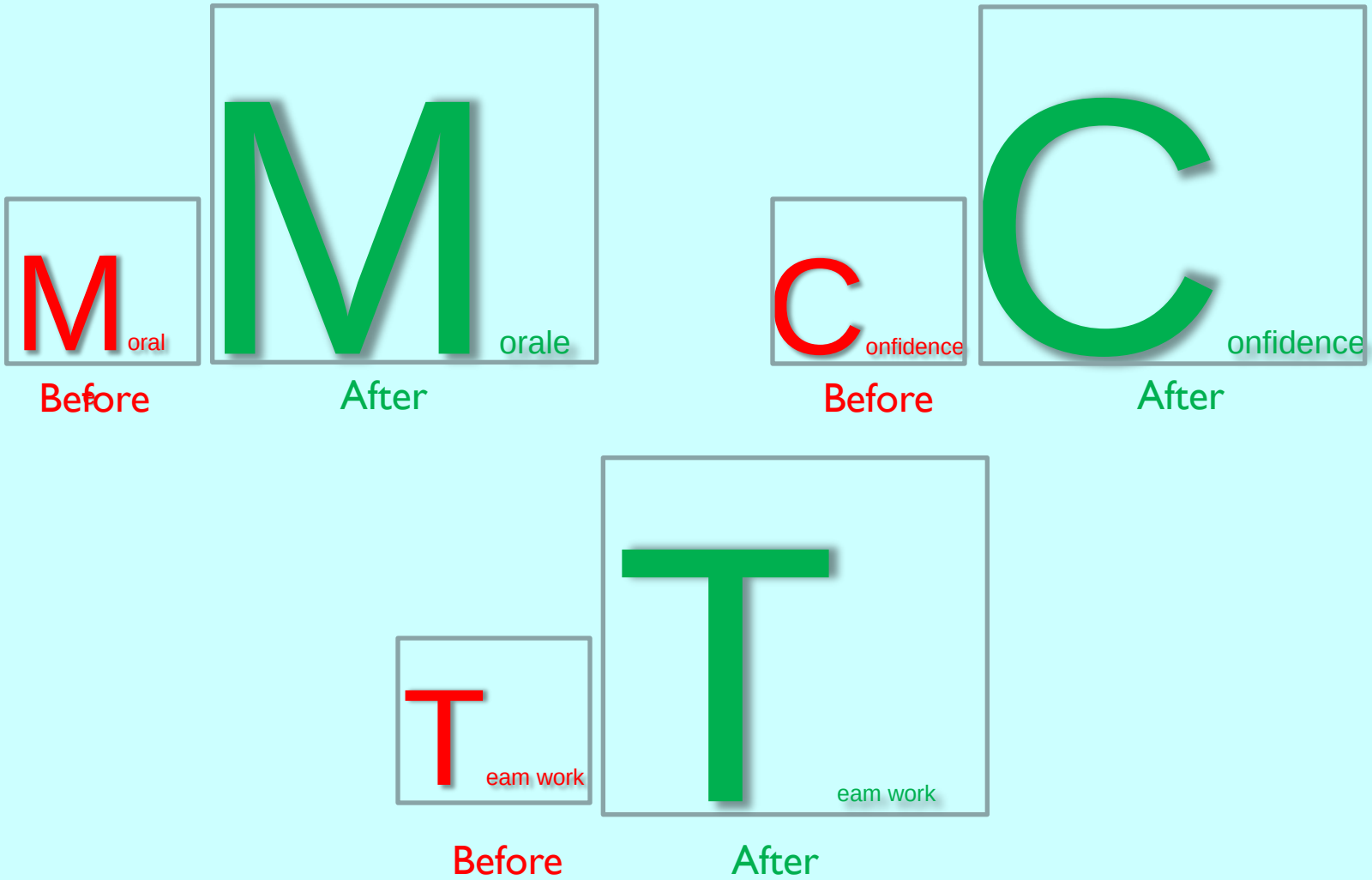
2.Defect and rework cost of Rs 2.73 Lac per annum is saved

3.ZERO bolt loose concern reported from RFI/PDI/CAI

4.No need to do the realignment of the back door

7.Check & confirm the results & effects (Cont……)

7.3 Intangible benefits:

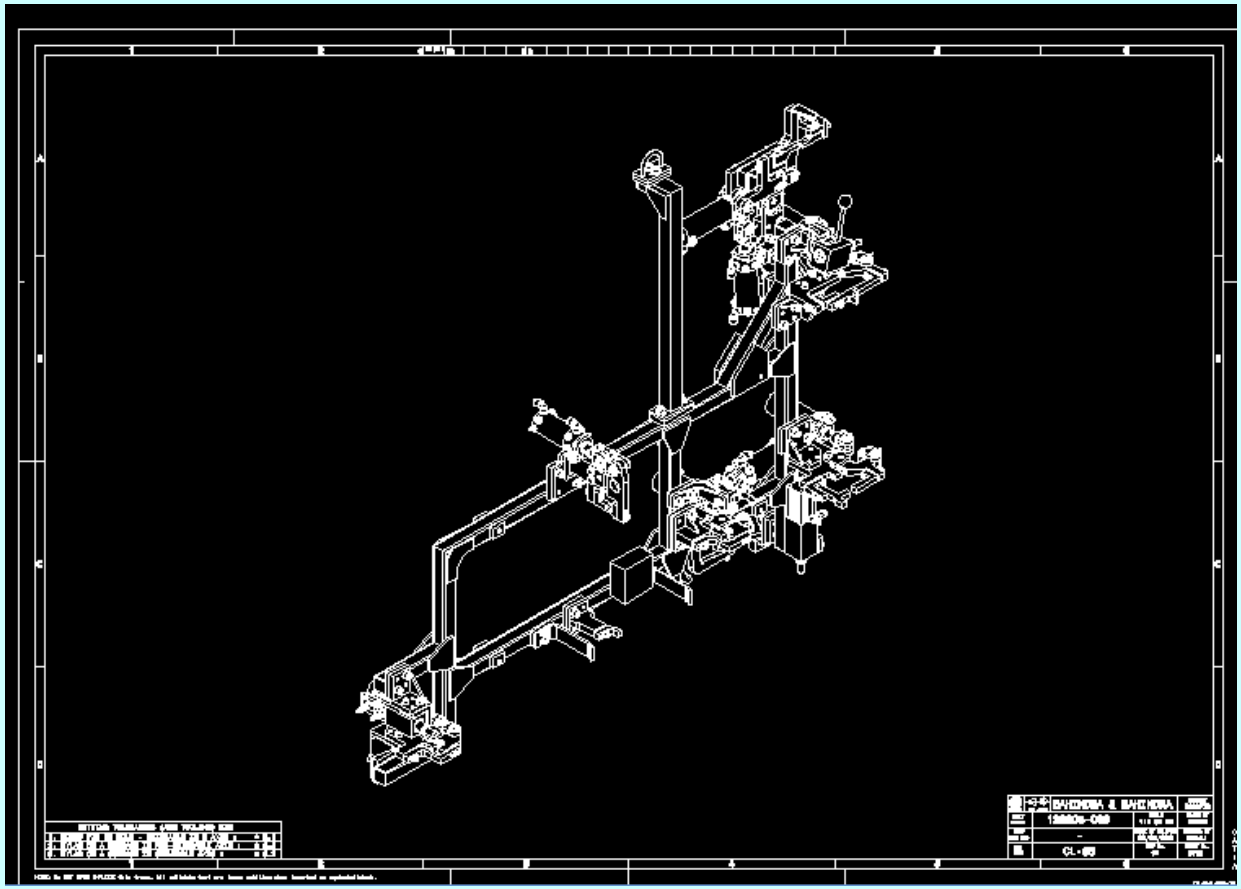


8. Standardization

8.1 Updation of documents:

Fixture Drawing Updated and revision no given on Tooling Record

Back Door setting fixture



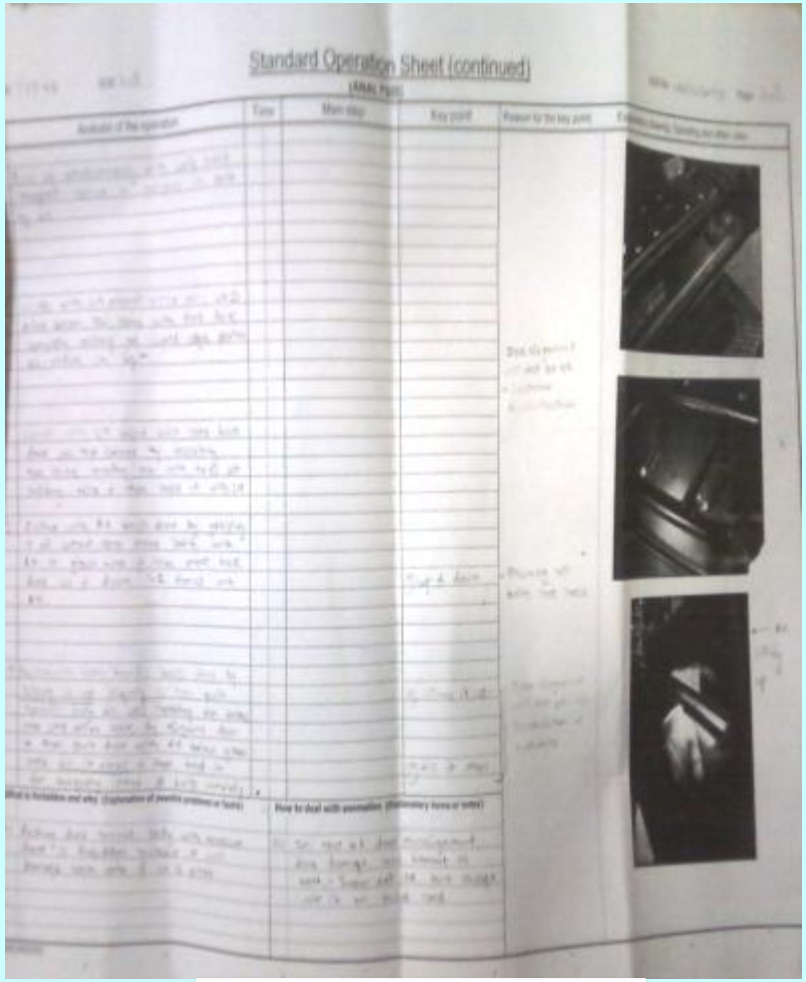
8.1 Updation of documents:

Fixture maintenance check sheet

	A	B	C	D	E	F	G
1	FIXTURE HISTORY CARD						
2	ZONE:	Z23				DEPT : MFG ENGG.-SCORPIO BODY	
3	FIXTURE N	Z23- 40		FREQUENCY : 6 Months and/or whenever changes done			
4	ASSY NAM	Back Door setting fixture					
5	DATE	UNIT NO	ITEM NO.	CONCERNS	CHANGES DONE/ ACTION TAKEN	REMARKS	
6							
7							
8	20/7/04			Back door alignment not ok	Back door setting gauge provided.		
9	30/10/05			Frame arm broken	frame arm welded with reinforcement		
10	15/12/05			To reduce fatigue for door fitment	Zero gravity balancer provided		
11	15/03/06			Fixture frame wornout & became weak	New frame installed.		
12	24/06/2007			Backdoor Missalignment	FRP fixture replaced		
13	5/9/07			Backdoor Missalignment	Magnetic spacer position is changed at RH side. Both magnet spacers to be placed		
14	10/2/2009			Back door bolt slippage issue	Back door closure fixture 2 mmshim added in +Z direction		
15	**18/4/09**			**Backdoor damage inside**	**Mgler profile chanded touching to**		
16	15/11/09			Wrong Fittment with marketing doors	Mechanical Pokayoke done to avoid wrong fittment.	i4-113974	
17	23/6/11			Damage Back Door outer panel	Balance regulator replaced	Scrap June11	
18	28/7/11			Backdoor damage inside	Provided guide rod with rubber hose for locating to avoid damage from locating pin.	CA2P0422	
19	18/4/12			Outer panel damage due to malfunctioning of peumatic clamp	Interlocking provided with clamp	i4-86025	
20	12/7/2012			Bolt slippage issue	Manual clamp converted into pneumatic connection.		
21							
22							
23							
24							
25							

8. Standardization

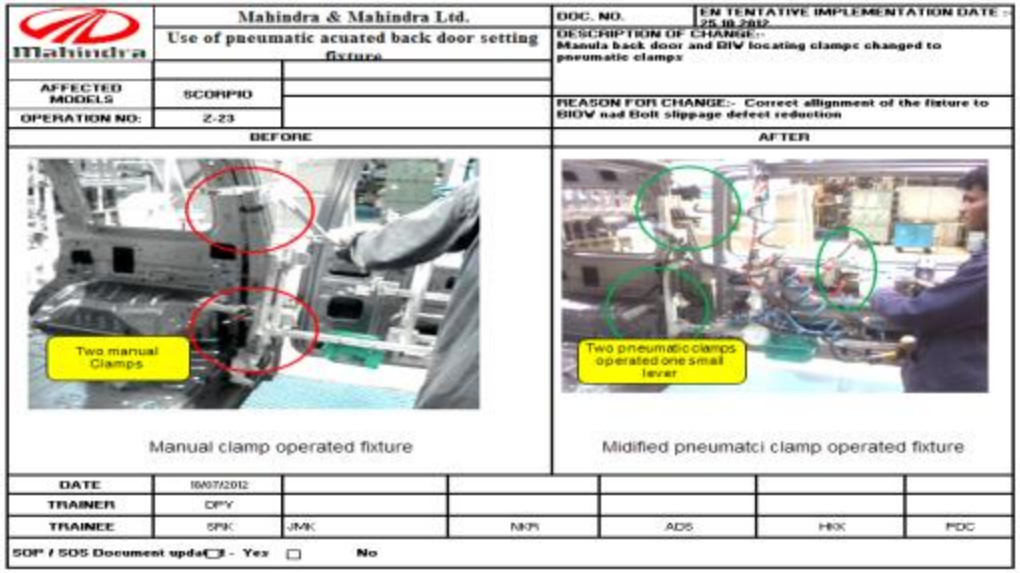
8.1 Updation of documents:



SOS

8.2 Training:

OPL Given to all the stage operators & **On Job Training** given to all the stage operators with the **I Do, We Do and You Do Technique.**



Mahindra & Mahindra Ltd.		DOC. NO.	EN TENTATIVE IMPLEMENTATION DATE :- 25.10.2012
Use of pneumatic acuated back door setting fixture		DESCRIPTION OF CHANGE:- Manula back door and BIW locating clamps changed to pneumatic clamps	
AFFECTED MODELS	SCORPIO	REASON FOR CHANGE:- Correct allignment of the fixture to BIOW nad Bolt slippage defect reduction	
OPERATION NO:	2-23		

BEFORE	AFTER
Two manual Clamps	Two pneumatic clamps operated one small lever
Manual clamp operated fixture	Midified pneumatci clamp operated fixture

DATE	18/07/2012					
TRAINER	DPY					
TRAINEE	SPK	JMK	NKR	ADS	HKK	PDC

SOP / SOS Document upda☐ - Yes ☐ No

8.3 Additional Audit / Check

100% checking at metal finishing stage for back door bolt loose and missing as a part of work and we are continuing it

9. Summarize & Plan future actions

9.1 Conclusion:

We have planned to make the defect reduction by Zero

Proposals1 : 1 Modification of RH D pillar Assy welding Fixture

Resp: CME Target DATE: Aug13

Proposals1 : Part Design change to make Hinge to nut and sleeve concentric in RH D pillar of BIW

Resp: PVT Target DATE: July 13

9. Summarize & Plan future actions

9.2 Horizontal Deployment:

We have already initiated the hood setting fixture auto clamping to avoid the bolt slippage

Responsibility: CME

Target DATE: JULY 13

9.3 Lessons Learnt:

1.Correct interpretation of collected data

2.Always challenge given inadequate process leading to innovative ideas thoughtfully implemented

QC Tools Used In Our Project

1. ESHIKAWA Diagrame
2. 5W 1H
3. Parato
4. Brain Mapping
5. Check sheet
6. Why Why Analysis

THANK YOU !!!